Analysis of Global Happiness

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***Abstract*—Recent years have witnessed a paradigm shift in measuring societal well-being, with happiness emerging as a critical indicator alongside traditional economic metrics. Leveraging data from the World Happiness Report, this study employs machine learning techniques to analyze the socio-economic determinants of happiness across 150+ countries. We investigate the impact of variables such as GDP per capita, social support, life expectancy, freedom, generosity, and corruption perception on national happiness scores.**

Our methodology compares multiple regression models (linear regression, Random Forest, Gradient Boosting, and Ridge regression) to predict happiness scores, with Random Forest achieving the highest predictive accuracy (R^2 = 0.849). Additionally, K-Means clustering identifies three distinct country groups based on happiness profiles, revealing nuanced patterns (e.g., high-GDP nations with low generosity). Key findings validate hypotheses linking GDP, health systems, and social support to happiness, while highlighting limitations in data granularity and cultural biases.

This research underscores the potential of data-driven approaches to inform public policy, emphasizing the need for holistic well-being strategies beyond economic growth. Future work could integrate cultural and individual-level data to enhance model interpretability and predictive power.

***Index Terms*—Happiness prediction, Machine Learning, Socio-economic factors, Random Forest, Clustering, Public Policy**

I. Introduction

Recent years have seen a fundamental change in the sense of happiness. Progressively , it is seen as an important worry for a society, just as major as aggressive crime rates. Apart from prototypal hints of economic rise including GDP, it gives a more nuanced evaluation of the welfare of a given population. Over the past 10 years, a World Happiness Report has been launched, estimating the level of happiness of citizens in over 150 countries once a year. It provides, among other things, the worth of social support, life expectancy, freedom, generosity and perceived corruption.

This precious data enables statistical analysis, as well as the application of current artificial intelligence aids,especially machine learning (ML) techniques, to model and predict happiness levels. Machine learning techniques, as an example, make it doable to evaluate the reciprocal consequence of each aspect and identify trends or groups of countries with homogeneous profiles.

International socio-economic states have been globally concealed by social researchers,aiming to estimate their effect on general happiness. I try to give to this extremely active research process by making use of the World Happiness Report dataset model. This guide to the following research questions: Which economic and social factors have the greatest impact on happiness levels worldwide?

Can a country's happiness score be predicted from these factors?

Are there groups of countries with more or less similar happiness profiles?

In answering these questions, we pose the following hypotheses:

*H1: Considering all other factors, GDP per capita is the most influential factor on happiness.

*H2: Countries with a well-functioning health system tend to have higher levels of happiness.

*H3: The support provided by society overshadows the impact of economic factors.

*H4: A country's happiness score can be effectively predicted using socio-economic variables via machine learning.

This research will rely on regression and clustering models to assess the importance of various predictors, estimate the happiness score and apply clustering to identify possible country groupings. The ultimate aim is to improve understanding of the mechanisms underlying perceived happiness and identify potential targets for public policy.

II. State of the art (existing work)

A. World happiness report

This report has one principal point : To suggest that national happiness should be a crucial objective for every government to follow. It looks to examine life pleasure over populations, concentrating on determining the drivers of well-being especially during hard times such as the COVID-19 pandemic. This article uses national surveys by which each individual is asked to rate their life satisfaction. It evaluates trust, kindness, altruism, and the functioning of state institutions. It also assesses social media data as a new way for calculating happiness and distress. Some of the most crucial conclusions are the growing agreement that life satisfaction evaluation should be the leading indicator of well-being, the importance of kindness together with benevolence in saving lives during a crisis, the relationship between nice governance and citizen welfare, and the impact of altruistic actions. Analysis of social media gives insight into the evaluation of global happiness. There are,yet , some limitations like life satisfaction evaluations tend to be

subjective, social media data can be influenced , and the factors that control well-being are complex.

B. Happiness and satisfaction -Our World in Data

The paper inspects differences in happiness and life satisfaction not only over nations and incomes, but also social variables, with the goal of reliably estimating the feelings across time and cultures. The research approves that increased income is typically associated with more life satisfaction, although its impact decreases with well-being according to worldwide surveys like the Gallup World Poll and the Cantril ladder scale. Nordic countries always rank as the happiest because of a mix of high earning, comprehensive social support, good governance, and individual freedom. Traumatic life events affect happiness in the short term, but people are likely to adapt—a process known as hedonic adaptation. The research shows that pleasure can be quantified and is strongly associated with objective indicators like mental health and longevity. A remarkable determinant of happiness are health, friendships, freedom, and confidence in institutions apart from money. In spite of some constraints—e.g., cultural variation and the fact that happiness may mean different things to different people—the article comes to an end with happiness being a real and quantifiable idea that public policy can target to increase general well-being, and not only economic advance.

C. Exploring the Reciprocal Relationships between Happiness and Life Satisfaction of Working Adults—Evidence from Abu Dhabi

The primary objective of this study is to explore the explore the reciprocal relationships between happiness and life satisfaction among working adults in abu dhabi . the research aims to examine how various well-being factors , such as job satisfaction, mental health, and social support , interact with these two subjective measures of well-being. To acheive this , the researchers utilized data from the second cycle of the abu dhabi quality of life (QoL)survey , which included 34,499 employed participants . The survey was administered online and in multiple languages , covering questions on various life aspects, including job satisfaction, work-life balance , and social relationships . the analysis of relationships between happiness , life satisfaction, and other variables was conducted using a path analysis model via the lisrel8 software. Results revealed a significant reciprocal relationship between happiness and life satisfaction, with direct effects from variables such as job satisfaction mental health , and satisfaction in personal relationships . in terms of model fit , the indices indicated excellent alignment (e.g.,$x^2/df = 1.79$). However , the study has certain limitations , including an over representation of public -sector workers in the sample, which complicate generalizing the findings to abu Dhabi's entire working population. Additionally , the use of cross-sectional data limits the ability to establish clear causal relationships between variables . In conclusion , this research demonstrates that happiness and life satisfaction are interconnected among abu dhabi workers , emphasizing the importance of social relationships and job satisfaction in enhancing overall well-being . it also highlights the need to consider contextual factors in workplace well-being policies

D. Machine learning for the analysis of quality of life using the World Happiness Index and Human Development Indicators

This article aims to identify the determinants of happiness at the international level by analyzing the links between objectives indicators (Human Development Index, HDI) and subjectives indicators (World Happiness Index , WHI) using statistical methods (Person correlation, principal component analysis) and machine learning algorithms (Random Forest, XGBoost, Decision Tree). using data from the World Happiness Report and Human Development Index (HDI) indicators spanning 2015-2021 across 119 nations , the study standardized the datasets and applied regression modeling . key findings highlight gross national income (GNI) PER CAPTIA as the most influential predictor of happiness . the random forest regression model demonstrated superior accuracy ,with an R^2 OF 0,93667 AND RMSE of 0,05748 outperforming closely trailing models like XHBoost.Additionally , a robust positive correlation between HDI scores and happiness levels was confirmed with Western European countries consistently ranking highest in both metrics However , the study has limitations : data bias (overrepresentation of developed countries, exclusion of 68 nations), inability to establish causality (cultural variables excluded), end excessive focus on economic indicators at the expense of social or environmental dimensions. despite these constraints , the ML approach uncovers complex interactions between varaibles , paving the way for future research integrating more diverse factors for a holistic understanding of well-being

III. METHODOLOGY

A. Description of dataset

1) Dataset
   Data7 : data of world happiness
2) Variables
   - Target: score of happiness
   - Features: PIB, soutien social, espérance de vie, liberté, générosité, corruption.
3) Key Observations
   a) Score Group Distribution
   - High: 23 countries (e.g., Nordic countries, Canada, Australia).
   - Medium: 80 countries (e.g., France, China, Brazil).
   - Low: 53 countries (e.g., Yemen, Afghanistan, Haiti).

   b) Trends and Patterns High-Score Countries:
   - GDP per capita (1.3)
   - Social support (1.4)
   - Healthy life expectancy (0.98)

   Notable Outliers:

- Generosity is paradoxically higher in some low/medium-score countries (e.g., Myanmar: 0.566).
- Corruption perception is low in Qatar (0.167) but high in Somalia (0.27).

4) Potential Analyses
   - Regression Analysis: Identify which variables most strongly predict the Score.
   - Clustering: Group countries with similar profiles (e.g., "High GDP but low generosity").
   - Geospatial Mapping: Visualize regional trends in ScoreGroup and key variables.
5) Limitations and Caveats
   - Normalization: Variables likely use an undisclosed normalization method (values range 0– 1.7).
   - Subjectivity: Metrics like "Freedom" or "Corruption" may reflect cultural biases.
   - Missing Countries: Key nations like Cuba or North Korea are absent.
   - Visualization Suggestions

   Heatmap: Show correlations between variables.
   This correlation analysis shows that GDP per capita (0.80), social support (0.76), and healthy life expectancy (0.76) are the factors most strongly related to national happiness. Conversely, generosity (0.11) has a marginal influence, while freedom (0.54) and perceptions of corruption (0.39) play a moderate role. Interactions between variables (such as the GDP/health link) mean that happiness depends on a complex balance between economic prosperity and social well-being, where wealth alone is not enough without strong support and health systems.
   Box Plots: Compare GDP per capita or life expectancy across ScoreGroup.
   Scatter Plots: Plot score versus social support (size points by GDP).
6) Pretreatment:
   - Replace missing values with the mean
   - Remove irrelevant columns
7) Models tested:
   - Split + Régression linéaire
   - Régression linéaire multiple
   - RandomForestRegressor
   - GradientBoostingRegressor
   - Ridge
8) Metrics:
   RMSE, R², importance of variables (feature importance). Please use "soft" (e.g., `\eqref{Eq}`) cross references instead of "hard" references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

IV. Results

A. regression

The R^2 (coefficient of determination) is a measure of the model's goodness-of-fit. It indicates what proportion of the variance in the target variable (e.g., happiness score) is explained by the model. The closer the R^2 is to 1, the better the model's ability to explain the variability in the data. The MSE (Mean Squared Error) measures the average squared error between the true values and the values predicted by the model. A lower MSE means that the predictions are closer to the true values. Finally, the RMSE (Root Mean Squared Error) is simply the square root of the MSE. It allows the mean error to be expressed in the same unit as the target variable, making it easier to interpret.

After testing various models, it appears that the RandomForestRegressor offers the best performance, with an R^2 score of 0.8492, indicating that it best explains the variance of the happiness score in the data. The improvement over traditional linear regression, which reaches an R^2 of 0.7487, is significant: about a 0.1 gain, which is far from negligible in a scientific project. This result shows that Random Forest is capable of capturing more complex relationships between explanatory variables and the target than simpler models such as linear regression or Ridge regression.

Regarding the complexity of the model, using a more complex model is justified here: in addition to a clear improvement in R^2, the mean squared error (MSE) also decreases (from 0.30 to 0.18). However, it is important to specify that while Random Forest is efficient, it is less interpretable than linear regression, as it is more difficult to explain precisely how each variable influences the prediction.

Finally, the choice of the model depends on the scientific objective of the project. If the main objective is to predict the happiness score accurately, then Random Forest is the model to prioritize. However, if the objective is rather to interpret and explain the effect of different variables on the score, a linear regression or Ridge regression would be more suitable, as they are simpler to analyze and present.

tested models	R^2 Score	MSE
Split + Linear Regression	0.7487	0.3083
RandomForestRegressor	0.8492	0.1849
GradientBoostingRegressor	0.8318	0.2064
Ridge	0.7489	0.3080
Multiple regression	0.7840	0.3080

TABLE I
COMPARING THE PERFORMANCE OF DIFFERENT MACHINE LEARNING MODELS

B. Clustering

The K-Means clustering algorithm was applied to the dataset, with the optimal number of clusters determined to be 3 using the elbow method. After clustering, the data points were assigned to one of the three clusters. A PCA (Principal Component Analysis) was performed to reduce the data to two dimensions for visualization, where the clusters were

displayed in a scatter plot. The mean values of the variables for each group were then calculated, providing insight into the characteristics of each group. This analysis reveals how the variables differentiate the clusters and helps to better understand the factors that influence the data.

V. Discussion

A. Interpretation of results

The results obtained in this study reveal that the RandomForestRegressor model outperforms other models in predicting happiness scores. With a coefficient of determination (R^2) of 0.8492, the model captures a large portion of the data variability and provides more accurate predictions compared to traditional linear regression, which has an R^2 of 0.7487. This difference of 0.1, although modest, is significant in a scientific context, as it suggests that Random Forest can better capture the complex relationships between explanatory variables and happiness scores.

Furthermore, the mean squared error (MSE) also decreased with Random Forest, from 0.30 to 0.18, indicating that the predictions are closer to the true values. However, it should be noted that, although the model performs better, it is less interpretable than linear regression. This means that it is more difficult to understand precisely how each variable affects the forecasts, which could limit its usefulness in applications requiring detailed interpretation of the results.

B. Validation of hypotheses

The hypotheses formulated at the beginning of the study appear to be largely confirmed. The first hypothesis, that complex models such as Random Forest would be more effective at explaining happiness scores than simpler models, is validated by the results. Indeed, the Random Forest model demonstrated superior ability to accurately predict, suggesting that nonlinear relationships between economic and social factors and happiness are better captured by more sophisticated models.

However, the second hypothesis, that linear or Ridge regression would be sufficient to interpret the effects of different variables, is partially validated. Although these models are simpler to interpret, their ability to accurately predict happiness is lower than that of Random Forest, which could suggest that simpler models do not fully capture the complexity of the relationships between variables.

C. Limitations of the study

This study has several limitations that should be considered. First, the data used are aggregated at the national level, which may mask significant variations at the individual or regional level. It is therefore possible that local factors or factors specific to certain demographic groups were not adequately captured.

Furthermore, some variables may be missing or underrepresented in the dataset, which could affect the quality of the predictions. For example, data relating to mental health, social inequalities, or traumatic events could be important factors in explaining happiness, but these elements are absent or poorly documented in the available data.

Finally, there is the risk of data bias. Happiness scores may be influenced by cultural perceptions or data collection methodologies that differ from one country to another. These biases could limit the generalizability of the results to other contexts or groups of countries.

D. Prospects for improvement

To improve this study, several avenues can be explored. First, integrating additional data, such as those from more granular sources (e.g., individual or regional surveys), could yield more detailed and nuanced results. Better accounting for missing or underrepresented variables, as well as implementing techniques for handling missing values, would also improve model quality.

Second, the use of more advanced models, such as deep learning (DLL), could offer even better performance by capturing more complex nonlinear relationships. However, this should be accompanied by a more in-depth analysis of interpretability to ensure that the results remain scientifically understandable and usable.

Finally, it would be interesting to explore the impact of qualitative variables, such as culture, religious beliefs, or social attitudes, on happiness scores, to broaden our understanding of the factors influencing overall well-being.

VI. CONCLUSION

1) Summary of contributions This study demonstrated that machine learning models, particularly RandomForestRegressor, offer superior prediction capacity for global happiness scores compared to simpler models such as linear regression. Thanks to its better ability to capture complex relationships between economic and social variables, Random Forest achieved an R^2 of 0.8492, confirming that nonlinear models are more suitable for predicting happiness scores. In comparison, simpler models, such as linear regression, showed lower performance. This work also identified groups of countries with similar characteristics through the K-Means algorithm, providing a clearer view of the factors influencing happiness in different regions.
2) Potential usefulness in public policies The findings of this study may have significant applications in public policy, particularly in the design of strategies to improve citizens' well-being and quality of life. By identifying the factors that most contribute to people's happiness, decision-makers can develop targeted policies to improve these aspects. For example, a policy focused on improving economic conditions or social equality could have a direct impact on improving happiness scores in certain countries. The use of data-driven predictive models can help anticipate future needs and measure the impact of implemented policies, thus facilitating more efficient management of public resources.

3) Openings: Although the study focused on economic and social variables, it is clear that other dimensions could enrich the results. The addition of cultural data (e.g., social values, religious beliefs, perceptions of happiness), individual data (mental health, social networks, relationship quality), and data related to local policies could help refine predictions and better understand the underlying mechanisms of happiness. For example, studies on the role of social cohesion or mental health policies could provide crucial insights.

Integrating data from regional or individual sources would also provide a more nuanced view of the factors affecting well-being, allowing for better targeting of interventions. Furthermore, exploring more complex models, such as those using artificial intelligence to analyze deeper interactions between variables, could provide an even more precise understanding of the social and economic dynamics influencing happiness.

In summary, this study opens the way to new avenues of investigation to better understand the factors that contribute to the well-being of populations on a global scale, while highlighting the importance of adapted models and varied data for richer and more precise analyses.

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